

DOCUQUERY:AN INTELLIGENT FILE-BASED CHATBOT USING VECTOR SEARCH AND NLP

A22126511040 - N.Jyotshna

A22126511059 - T.Pragna

A22126511045 - P.Sravani

A22126511051 - S.Sri Ram

Under the Guidance

of

Mrs.D.Tejaswi ,Assistant Professor

Department of Information Technology



ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY & SCIENCES(Autonomous)



Abstract

In today's digital age, professionals in fields such as law, education, and enterprise management face challenges in extracting relevant information from large volumes of documents, as traditional keyword-based search often fails to provide accurate, context-aware results. To address this issue, the project titled **DocuQuery: An Intelligent File-Based Chatbot Using Vector Search and Natural Language Processing (NLP)** aims to develop a system that allows users to query documents in natural language and receive precise, contextually relevant responses. The methodology integrates embedding models such as Bidirectional Encoder Representations from Transformers (BERT), Sentence-BERT, and OpenAI embeddings to convert text into semantic vector representations, combined with Facebook AI Similarity Search (Faiss) and cosine similarity to retrieve relevant sections. Large Language Models (LLMs) including Generative Pre-trained Transformer 4 (GPT-4) and Text-to-Text Transfer Transformer (T5) generate human-like answers, while Optical Character Recognition (OCR) ensures compatibility with image-based documents. The system supports multiple formats such as Portable Document Format (PDF), Microsoft Word Document (DOCX), and plain Text (TXT), with intelligent preprocessing to enhance retrieval accuracy. Backend integration is achieved using Python-based Application Programming Interfaces (APIs) like FastAPI or Flask, and a user-friendly web interface enables seamless interactions. The findings show that DocuQuery delivers more accurate and versatile document query resolution compared to existing systems like InfoGenie, which face restrictions in file type support and contextual understanding. This solution enhances accessibility, improves efficiency in information retrieval, and demonstrates commercialization potential as a Software-as-a-Service (SaaS) for legal firms, universities, research institutions, and enterprises. Overall, DocuQuery represents a scalable technological innovation designed to address real-world challenges in intelligent information management.

18-09-2025

ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY &
SCIENCES(Autonomous)



Outline

- 1.Introduction
- 2.Literature Survey(in table format)
- 3.Existing System with limitations
- 4.Proposed Methodology with advantages
- 5.System Architecture
- 6.Conclusion and References



Introduction

- In sectors such as law, education, and enterprise management, professionals face challenges in extracting precise information from large volumes of documents, as traditional keyword-based search fails to provide accurate, context-aware results, causing inefficiency and reduced productivity.
- Advances in Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) enable context-aware systems that use semantic embeddings and vector search to retrieve relevant information, while Cloud deployment ensures scalability and accessibility.
- This project, **DocuQuery: An Intelligent File-Based Chatbot Using Vector Search and Natural Language Processing (NLP)**, converts documents into embeddings using models such as Bidirectional Encoder Representations from Transformers (BERT), Sentence-BERT, and OpenAI embeddings, retrieves results via Facebook AI Similarity Search (Faiss), and generates human-like answers with Large Language Models (LLMs) such as Generative Pre-trained Transformer 4 (GPT-4) and Text-to-Text Transfer Transformer (T5), with Optical Character Recognition (OCR) enabling support for image-based files.
- A Flask or FastAPI backend and a React JavaScript (JS) frontend provide a user-friendly interface for document upload and interaction, supporting multiple formats such as Portable Document Format (PDF), Microsoft Word Document (DOCX), and Text (TXT). Compared to existing tools like InfoGenie, DocuQuery delivers more accurate, scalable, and versatile query resolution, with strong commercialization potential as a Software-as-a-Service (SaaS) solution for legal, academic, and enterprise sectors.



Literature Survey(in table format)

Author (Year)	Topic / Title (short)	Methods / Approach	Dataset / Domain	Key Findings / Project Link
Dik L. Lee et al. (1997)	Document Ranking and the Vector-Space Model	Cosine similarity, tf-idf weighting, relevance feedback	CACM, TREC datasets (text retrieval)	Approximate normalization improves efficiency; relevance feedback boosts retrieval precision.
Yu-Chuan Chang et al. (2006)	Query Reweighting via Genetic Algorithms	Genetic algorithm encoding for query terms	User query logs, document corpora	Genetic optimization enhances recall and precision over baseline retrieval methods.
Kukreja et al. (2025)	Performance Evaluation of Vector Embeddings (RAG)	Various embedding models, ensemble techniques	NCERT education corpus, RAG framework	Ensemble embeddings enable faster retrieval; individual models improve semantic quality.
Ratna Patil et al. (2025)	Intelligent PDF Query System for Document Analysis	AI-based PDF parsing, FAISS, transformer models	Mixed PDF datasets (technical, academic docs)	NLP transformers + FAISS enable accurate, rapid PDF querying (87% accuracy).
Salima Lamsiyah et al. (2023)	Query-Focused Multi-Doc Summarization (Anaphora Resolution)	SpanBERT for pronoun resolution, SBERT embeddings, MMR	DUC2005–2007 multi-doc news collections	Anaphora resolution and contextual embeddings improve summary coherence and retrieval relevance.
Hoang & Nguyen (2025)	Multimodal PDF Retrieval Augmented QA	DBSCAN cleaning, chunk embedding, RAPTOR, LLM fine-tuning	Financial/engineering PDFs, mixed document types	RAG-aware QA system handles multimodal retrieval (text, tables, images) with high accuracy.

SCIENCE(Autonomous)

Existing System with limitations

Most existing document-based query systems rely on traditional keyword search, statistical vector methods, or limited-format parsing. These approaches provide some basic retrieval capability but face several challenges when applied to real-world, large-scale document collections.

- Complex Document Extraction**

Struggle with scanned PDFs, tables, and multi-column layouts due to weak OCR and lack of layout analysis.

- Semantic Understanding Gaps**

Keyword-based and TF-IDF methods fail to capture true meaning, natural-language queries, and multi-step reasoning.

- Manual Feedback Dependency**

Require explicit user feedback and manual term selection, making them less automated and unsuitable for conversational use.

- Scalability Issues**

Not optimized for large or diverse datasets; slow response and reduced accuracy with many users.

- Citation & Traceability Limitations**

Provide plain text answers without references, making verification difficult and reducing trust.



Base Paper Reference and Analysis

- **A New Query Reweighting Method for Document Retrieval Based on Genetic Algorithms (Wang et al.)**

This paper supports the use of **optimization techniques** by showing how genetic algorithms can improve query reweighting, leading to more accurate document retrieval.

- **Can Anaphora Resolution Improve Extractive Query-Focused Multi-Document Summarization (Raghunathan et al.)**

This study demonstrates how **anaphora resolution** enhances query-focused summarization, validating the role of **context-aware NLP** in improving retrieval quality.

- **Intelligent PDF Query System for Document Analysis (Patil et al., 2025)**

This paper shows the value of **AI-driven PDF querying** using transformers and FAISS, justifying advanced architectures for **intelligent document interaction**.

- **PDF Retrieval Augmented Question Answering (Hoang & Nguyen, 2025)**

This work extends **RAG** to handle text, tables, and images in PDFs, supporting the development of **multimodal retrieval systems**.

- **Performance Evaluation of Vector Embeddings with Retrieval-Augmented Generation (Kukreja et al., 2024)**

This study compares **vector embeddings** within RAG, providing insights into which models give better retrieval and generation accuracy



Proposed Methodology with advantages

•Multi-format Ingestion & Preprocessing

Supports PDF, DOCX, TXT, and scanned documents with Optical Character Recognition (OCR) and intelligent chunking.

Advantage: Expands usability across diverse document types and improves retrieval accuracy.

•Semantic Embeddings & Vector Search

Uses models like BERT, Sentence-BERT, and OpenAI embeddings with Faiss and cosine similarity.

Advantage: Provides context-aware, precise, and relevant search beyond simple keywords.

•Response Generation with Large Language Models (LLMs)

Integrates GPT-4 and T5 for human-like, context-rich answers.

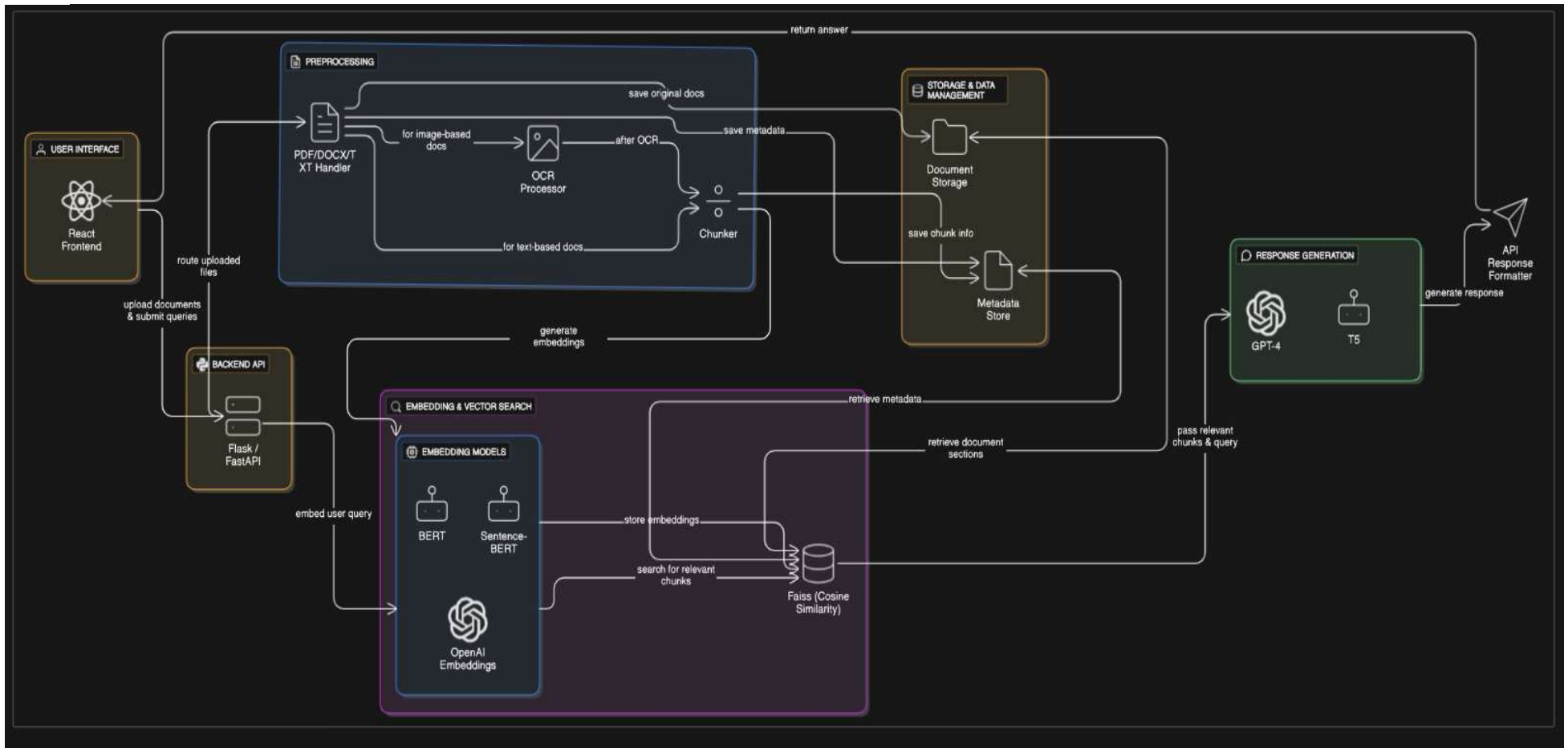
Advantage: Delivers natural, conversational responses suitable for multiple domains.

•Integration & Traceability

Backend with Flask/FastAPI and frontend with React JS, plus citation-backed outputs and scalable modular design.

Advantage: Ensures seamless interaction, transparency, and adaptability for enterprise use.

System Architecture



Conclusion and References

- The integration of Natural Language Processing (NLP), semantic embeddings, and vector search in DocuQuery marks a major shift from traditional keyword-based search to intelligent, context-aware document retrieval. By using advanced models such as Bidirectional Encoder Representations from Transformers (BERT), Sentence-BERT, and Generative Pre-trained Transformer 4 (GPT-4), along with Facebook AI Similarity Search (Faiss) and Optical Character Recognition (OCR), the system achieves high accuracy while supporting multiple formats like Portable Document Format (PDF), Microsoft Word Document (DOCX), and Text (TXT).
- DocuQuery's modular design ensures seamless deployment across legal, academic, and enterprise domains, offering scalability and adaptability. Unlike existing tools such as InfoGenie, it provides personalized, context-rich responses that improve knowledge management and accessibility.
- Ultimately, DocuQuery transforms unstructured data into actionable insights, reducing manual review time and enhancing productivity. With strong commercialization potential as a Software-as-a-Service (SaaS), it stands as a scalable and innovative solution for intelligent information management.

Thank You

18-09-2025

ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY &
SCIENCES(Autonomous)